

Reviewer Comments

We thank the reviewer for their thorough and detailed analysis of our paper. We highlight any changes and additions in the draft in bold. We also address each of their comments below; our responses are in bold and bulleted.

General comments:

Some of the data that is mentioned/described in this paper will be presented in future works, e.g., the radio data of AT2024qfm, and this is not made clear in the text why this is. The paper is presented as a full analysis of all the data for this sample of six LFBOTs, but some of the data is missing this way, leaving gaps in the dataset and this could potentially influence the results of this work, especially since some of the conclusions are based on the radio data. It needs to be explained why not all of the observations are included in this work and how this influences the results and conclusions.

- **We add lines clarifying that this was the result of prior arrangements with other research groups, and that the data deferred to future publications does not significantly influence the results and conclusions**

On that same note, AT2024aehp is mentioned as a peculiar LFBOT throughout the manuscript. It might be worth thinking about writing a shorter paper on this source as it is even suggested at the end that this source might have a different progenitor system than the rest of the sources. If the source stays included in the manuscript, it would benefit the structure if all of the discussion on AT2024aehp is combined in one section, so that all the information is together, rather than describing its peculiar behaviour spread out through the paper.

- **We agree, and do have plans to write a paper focusing on AT2024aehp, which will include new radio data that has been taken since the writing of the manuscript (this is already stated at the end of Section A.6). We collect the discussion on this transient's uniqueness into a subsection in the discussion.**

The structure of the paper could be improved in certain sections. The radio analysis should be moved forward to go directly after the observations, because it is analysis of work done. The host galaxy properties section should be its own section and should be closer to the discussion section.

- **Thank you for these suggestions. We move the Progenitor Discussion section to the wider Discussion section, and split the analysis into radio analysis and host-galaxy analysis**

Currently, not a lot of conclusions are drawn in the results and discussion sections, or in the summary. While it is clear that doubling the sample size of LFBOTs is impressive, especially considering the events were discovered in real time and could therefore be follow-up while the events were evolving, the implications of doing this is left up to the reader to figure out.

In the results and discussion section, a lot of things are implied rather than stated explicitly. While reading these sections, the reader is left wondering what certain statements mean in the context of the progenitors of LFBOTs and the impact of the presented observations on the general field, and the manuscript would benefit significantly if those discussions would be added. The content is all there, it just needs to be written out explicitly.

- **We expand on the details of the results and discussion sections, with a focus on the specific points brought up in the relevant sections of the provided feedback.**

As a minor point, most of the figures have the names of the LFBOTs written inside the plot, often covering a lot of the background data also presented. This makes it difficult to see the background information that is there for comparison and used to draw conclusions. Perhaps it would be possible for the labels to be included in the legends if they are blocking important data as is (e.g., in figure 3 and figure 7)

- **Thank you for the suggestion—we add the labels to the legend itself for these figures.**

Figure 1: The way the axis in the middle overlap makes this plot confusing. Splitting the columns and making the right-side axis labelled with 'Apparent AB mag' on all panels a lighter grey and including the left-side axis also on the panels in the right column to prevent confusion between right-side axis of the left column and the left-side axis on the right column would improve this plot. The caption has two mentions of a dashed line, which is confusing when reading for the first time or for colour blind people. Making either of these lines a different style removes any ambiguity about which line is which.

- **We implement these changes, including the change of the vertical lines to dotted lines, adding labels on all axes, and changing the apparent Mag axes to a gray.**

Figure 2: The caption of this figure mentions light and dark lines, which corresponds to binned and unbinned spectra, but there is only one line for each spectrum in the plot. This should be changed to reflect what is actually shown. A comment should be added in the text or the caption about the confidence level at which the Ca H+K lines are detected, because from the figure they could just be noise. If the H α line in the AT2023vth SEDM +2 days spectrum was not used for the redshift determination, which seems unlikely based on the lack of features in this spectrum, it should not be shaded. Would it be possible to denote in some way for which spectra it is assumed that the transients is contributing significantly and in which the light is dominated by the host galaxy? This will aid in demonstrating the timescales for these events.

- **Some lines like [N II] and [S II] are galactic lines and indicate a host-galaxy contribution. There is another factor to the timing that determines whether the host-galaxy contributes, which is the size and positioning of the aperture used to produce the spectrum. We add text around Line 212 describing this. We fix the error in the caption, and remove the H α shading in the SEDM spectrum.**

Figure 3: In the caption the labels used in the legend of the plot should be defined. Different line styles should be used for LGRBs and LLGRBs, so even in black and white it is clear what data corresponds to which subclass of GRBs.

- **We fix the caption to directly define LGRBs and LLGRBs, and change their line styles**

Figure 5, the colours need to be more different between the different epochs. In the panel for AT2023vth the markers are almost impossible to distinguish, both in colour and in markerstyle. Perhaps outlining all the markers with black and removing them for the previous observations of AT2023fhn would help already.

- **Thank you for the suggestion, we outlined the markers in black, and slightly adjusted the colour scale to try to emphasize the contrast**

Figure 7 has too much information on it. All the text makes it difficult to read. Perhaps leaving out the times will solve this already. Outlining the markers in black will help with clarity and helps distinguish between the observations in this paper and the background LFBOT data.

- **We remove the times and move the object labels to the legend**

Figure 8: Is there any way to show errors on the distributions of the offsets to the hosts in this plot? Without errors it is difficult to make any real statements about how consistent distributions are.

- **We do not have errors on the offsets of most past LFBOTs, as well as for the data for the large samples of other transients. Even if we found errors for all objects of a given sample, it would be hard to translate the collection of errors in each individual offset into an error for the cumulative distribution, especially in a readable way given that we plot many different distributions in this figure.**

Figure 11: Either the top or the bottom panel for each population could include the p-value of the KS test that is currently presented in Table 3, which would mean Table 3 can be removed from the paper. This would save space and including the p-value in the figure would make the figure clearer in showing how consistent (or not consistent) each population is with the LFBOT population.

- **This is a great idea; we have made these changes, and adjust the text and figure caption accordingly**

Figure 10: increasing the marker size for the observed LFBOTs will help emphasize this data within the larger sample.

- **We make these slightly bigger. We do want to keep the error bars visible for some of the points, so we have an upper limit on the size.**

Abstract

A large part of the abstract focuses on the one outlier observed in this paper, while a large part of the body of the paper is spent comparing the expanded sample of LFBOTs to itself and other populations. Therefore the abstract in its current form does not reflect the content of the paper

well enough and the paper would benefit if the abstract was rewritten to better reflect the content of the paper.

- **We shorten the description of the one outlier LFBOT, as it is not the paper's main focus**

Line 49: Is a radio or X-ray observations required, or can a source also have a detection in both? In that case this should be clear from the text. From the abstract it is unclear why the specific selection criteria were used as well.

- **We change the wording to and/or**

Line 59: This is the first and only mention of the LFBOTs in this paper being optically selected, this should be mentioned earlier in the abstract and also explained in the introduction. If there is no other selection method for LFBOTs, 'optically selected' can be omitted entirely.

- **By optically selected, we were referring to our use of the optical light curve for initial classification of an LFBOT candidate. As we only require a radio detection, and the other constraints for an LFBOT classification are in the optical band, we would not expect our classification criteria to enforce a strong bias towards making the radio behavior similar for all LFBOTs. As we don't define this anywhere in the text and we do not have much room in the abstract, we omit the phrase 'optically selected.'**

Line 59-60: It is unclear from the abstract alone why similar circumburst medium for each event implies the progenitor scenario posed.

- **Our logic was intended to go as follows: we infer a similar circumburst medium for each event -> the progenitor scenario must be able to produce a consistently similar circumburst medium -> one such scenario would be a merger. As explaining why a merger could produce a consistent medium would require more detail than we have space in the abstract, we have edited the abstract to omit the last step and leave it for the discussion section to explain.**

Introduction

General

The introduction is concise and clearly indicates the wide spread of models suggested for LFBOTs, however, it does not introduce the already known population of LFBOTs in enough detail. While this paper is clearly focussing on the group of LFBOTs in general, there is a peculiar one discussed in detail later in the paper. From the introduction it is unclear that the group of LFBOTs is quite heterogeneous already (some works suggest that this could be due to viewing angle, e.g., Chrimes et al, 2024, A&A, Volume 691, id.A329), and so finding one with properties different to the rest of the sample is not completely unexpected. It would be good to add a paragraph on how some of the other LFBOTs also show peculiar characteristics (e.g., 18cow shows this long term UV plateau, Sun et al., 2022, MNRAS: Letters, Volume 512, Issue 1, Inkenhaag et al, 2023, MNRAS, Volume 525, Issue 3) This way it is easier to argue why AT2024aehp can still be considered an LFBOT, despite potentially having a different progenitor.

- **We have added these references, and added a paragraph discussing how past LFBOTs do often have characteristics unique to themselves**

For clarity, it should be stated in the introduction what timescale this paper is interested in, from the rest of the paper it becomes clear that this paper focuses on early time follow-up, but despite this, it would be good to quickly discuss the late-time properties of the known sample of LFBOTs, and then stating that this paper only focussed on early time observations (state till what day roughly as well).

- **We make a mention that the paper presents observations up to 150 to 400 days after discovery (depending on the object). The late-time properties are still actively being investigated as LFBOTs are relatively recent discoveries, but we cite papers discussing these.**

Specific

L69: 'Some transients' → 'Some of these transients'

L76: 'FBOT' → 'FBOTs'

- **Thank you for the corrections, we make these specific changes.**

L72: $g-r \leq -0.2$ is different from $g-r \leq -0.5$ mentioned in the abstract, these should be consistent

- **We check our sources, and -0.2 is the cited value, so we amend the -0.5 to match this.**

L77: If the class of FBOTs that is focussed on is specifically required to have an X-ray and/or radio observation, they are more X-ray/radio selected than optically selected like mentioned in the introduction. Could sources be missing from the sample because of this specific selection criterium? This needs to be discussed either in the introduction or in the discussion.

- **We add a summary of the detection criteria in the introduction. In the discussion subsection for selection biases, we add a statement that given a promising LFBOT candidate in the optical, we often find X-rays or radio emission with a low failure rate, so we believe that the X-ray/radio requirement does**

L81: Are all these 8 sources indeed confirmed to be LFBOTs, or are some of these still speculation to fit in with this castigatory?

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Line 120: 'LFBOT' instead of just 'FBOT'.

- **Thank you for catching this, we have made this correction.**

Line 123: "In this paper, .." should be a new paragraph that describes the novelty of this manuscript, this can be merged with the next paragraph, so there is one paragraph describing the new results and the work of this paper.

- **We have separated this into a new paragraph. We do keep the next paragraph separate, as it does contain the specific section-by-section summary of the paper.**

L 128: Why is there no UV data for any of these sources? Is that because of a lack of time on a relevant telescope or was this a strategic decision?

- **For Swift UVOT data, only AT2024qfm and AT2024aehp had a good source model to successfully perform source subtraction. We have added the AT2024qfm UVOT points, as the AT2024aehp UVOT was already present.**

Much appreciated that AB mags are clearly indicated.

Observations

As is quite often the case with sections describing observations, this section is quite lengthy and dry. Using Tables to list the observations does minimize this, which is appreciated. If possible, having a panel with images of the six sources on the sky would make this section a bit more interesting. These should indicate the location of the transient and show which galaxy is its (most likely) host galaxy. That way it becomes more intuitive when talking about the host galaxies later on in the paper. This is of course only possible if such data exists, but can be archival data.

- **We refactor the description of the spectroscopy into the spectroscopy table in the appendix, and add host galaxy cutouts as Figure 7**

The bottom row in Table 1 on page 4 is empty, apart from the LFBOT name, and needs to be filled in or explained why it is empty. The redshift presented in this same table does not contain an error. Even if the error is small, or just in the last decimal, this should be mentioned.

- **The bottom two rows refer to the same transient. We re-position some of the words to make this clearer. We add a note that the redshift error is to the last reported decimal.**

Section 2.1

L169: A little more detail could be provided on how the MW extinction was applied (built into the pipelines, python package, etc.)

- **We add the detail that we used the Python extinction package**

L173-174: Not all LFBOTs will have enough data points to get an accurate estimate for the peak of the optical light curve, so an explanation of how the peak of the light curve was determined is needed. Even if it is obtained from other work/ATELs/GCNs, this should be stated.

- **For simplicity, we just use the time of the brightest optical detection as the location for the peak. As you pointed out, this does incur one-day margin-of-error, but we deem that to be acceptable for our analysis**

Section 2.2

This is the only subsection in the observations section that includes lengthy descriptions of the spectroscopic observations of all sources, with all instruments, setups, exposures, etc. written out. Turning this section into a table in the appendix would shorten the body of the paper and

improves clarity on all the observations done by having them all in one place. Another argument to turn it into a table is so you can make sure no information is left out, such as the standard star used for the different spectra calibrations, which is not mentioned for every source currently.

- **We agree that the table is best suited for this information. We remove the redshift and column on detected lines from the table as this can be determined from the raw data (which we plan to provide) and the plot. Some information is omitted, but it is info that we do not have for all observations anyway.**

L229: It should be described how the most likely host galaxy for each source was determined, especially for previously unpublished sources.

- **For this sentence, we add that the most likely host was the one closest to the transient. We add a brief description of how we determine the host galaxy for each LFBOT in the paper at the start of Section 4.2.**

L244: While it is appreciated that the github link to the codes is provided, a short description of what these codes do would make the process clearer for non-experts.

- **We provide a short description of the codes in the table comments.**

L253: For AT2023vth the standard star used is not mentioned.

- **As we do not have standard star information for all observations, we remove the mentions of standard stars.**

L260: DRAGONS is mentioned here again, but neither here nor when it is previously mentioned is it explained what this code does and why this specific code is used for this source/data in particular.

- **We add a sentence in the table footnote that DRAGONS is specialized for reducing data from the Gemini Observatory.**

L264-265: Are the central wavelengths mentioned here correct? These are very close to each other and it is unclear to me why two observations with central wavelengths so close are needed.

- **Good question—the GMOS detectors have gaps, and a small dither of about 10 nm (100 Angstroms) ensures that the spectra is continuous. We remove the details of central wavelengths as it can be determined from the displayed spectra and data if needed.**

L268: A description of what a 'classical night' would aid in understanding why this is important.

- **We remove the mention of a 'classical night'.**

L283-284: The zenith/airmass should be mentioned for all source, or for none.

- **This data was not taken for all sources, so we removed all mentions**

L301-308: The data reduction procedure for these observation are described in more detail than for other sources. This should be consistent between the different observations, perhaps the software/packages can be a column in the big table.

- **We agree, and added a software column in the table. Brief descriptions of the software are provided in the table comments**

Section 2.3

This section is nice and concise for the amount of X-ray observations presented. The description of the check on the detection of emission for AT2024qfm is appreciated.

L353: 'a factor of a few', this is too vague. It is unclear how significant this result is based on this phrasing.

- **Changed to a factor of three. Most of the difference is likely from the fact that they bin the latter two observations together, causing it to be a deeper upper limit**

L354-356: Chandra is a different telescope than Swift, so a comment needs to be added on if and how we can directly compare the data from AT2023fhn to the Swift data of the new LFBOTs.

- **Added a comment that Chandra measures flux over a very similar energy range (0.2-10 keV vs 0.3-10 keV), so the measurements should be very comparable.**

L358: five of the new LFBOTs or five of the whole sample of LFBOTs?

- **Five of the new, added the clarification**

Section 2.4

L373-374: An explanation of why the radio data of the sixth source will be presented in another paper? It makes the current dataset incomplete and therefore weaker.

- **This was an arrangement that we made with another group. The set of five additional radio light curves already increases the sample size of published LFBOTs with radio observations by ~50%, so we do not feel that this takes away from the paper in any significant way.**

L377-378: 'Based on past behaviour...this band', this needs a reference to previously published radio observations of LFBOTs from which you draw this conclusion of needing to observe all sources at 10GHz.

- **We add references to the observations of AT2020mrf (Yao et al), AT2020xnd (Ho et al), and AT2023fhn (Chrimes et al). Emission at 10 GHz would be visible for a prolonged time window as the radio SED would peak just above 10 GHz, and then slowly move lower so the peak passes through the 10 GHz range.**

L420-422: A clearer explanation of this error is needed. Is this precision on the flux scale a systematic error, or is this a statistical error that is the same in each band regardless of the brightness of the objects?

- **We add a statement that the precision limit of the flux scale is a systematic error**

Section 2.5

L460: The limits used when determining if the localisation of a GRB is consistent with the locations of the LFBOTs should be stated.

- **The GCNs report localizations and errors, and we check if the LFBOT is within 3-sigma. We add the 3-sigma note to the paper.**

Section 2.6

This section reads like the work was only done to confirm the Metzger model, and confirmation bias might have crept in. It is definitely a good test to do, trying to find X-ray precursors, but either a justification of why only this model was the motivation to do this work should be added, or more work needs to be done doing the same work for other models (this could be in other wavelengths as well).

- **We are not aware of another model that predicts X-ray precursor emission. We added a sentence to clarify this.**

L478: Clarify if consistency in location is the first step, only after this the time is taken into account.

- **Each field is imaged once as part of the all-sky survey of the Data Release, so we only check for location consistency, and accept any possible time for the observation.**

L486: The distance used to determine consistency between a detection of the location of an LFBOT should be stated.

- **As stated in the text, the closest catalogued sources were several arcminutes away. We added text to clarify that this is far greater than the spatial resolution of eROSITA (~ 10 arcsec, Predehl et al. 2021).**

Section 2.7

L512: Discuss how using a different SNR threshold would influence the amount of datapoints included and how that would influence the results.

Added a couple of sentences on the choice of SNR. It was chosen to try to keep detections in our data set, to the point where lowering the threshold further would have diminishing returns.

L529: A more detailed description of why this particular aperture size was chosen and why it is the same for all sources except when a bright source is detected nearby should be added.

- **Added a couple sentences explaining our reasoning to use this aperture size, and that we hoped to keep it as consistent as possible, except for the host galaxy's with nearby sources that required a smaller annulus to exclude nearby source flux from the background.**

Section 3

This is where the structure of the paper starts to break down. Sections 3-5 mix results with conclusions and discussion. While that is not necessarily wrong, and with the amount of information in this paper could make a lot of sense, it would require more structure within the

subsections. Some statements made in this section need to be expanded on in the discussion as well.

Section 3 feels like it should be moved to the discussion, as it discussed the results from the different wavelength observing campaigns in the context of comparing it to other transients and the full LFBOT sample, it would then also naturally allow for more discussion of results in context of the progenitor models.

- **We agree and move this section to the discussion. We also move the mentions of AT2024aehp into its own section in the discussion, in keeping with comments made on the old Section 5.**

L539-543: This sentence states that the objects were identified on the basis of fast rise and decay, but the next sentence says the rapid decay was not selected for. This contradicts each other and needs to be clarified.

- **Clarified that the fast rise and decay that we select for is within the first few days of discovery and optical peak. The continued decay past ten days is not selected for, as evidenced by AT2024aehp being classified as an LFBOT while having a late optical plateau.**

L543: Can we also rule out the plateau phase observed in AT2024aehp in the sources that aren't explicitly mentioned here?

- **No, unfortunately we did not do deep enough optical follow-up in the early LFBOTs as we did not account for a plateau happening. We add text explicitly mentioning this again in Section 5.2, part Observations, as an avenue for improvement**

L547-550: The language used in these sentences is too vague. 'fairly similar across most of the sample except for AT2024aehp' is not exact enough. Remove 'fairly', and if possible add a quantitative measure of how consistent the radio behaviour is across the sample. Also remove 'most of the sample', the exception is already mentioned, so no need to add 'most of the'.

- **We remove these instances of redundant language. The further sentences are meant to describe how the radio evolution is similar, taking into account peak luminosity, peak frequency, and SED shape.**

L554-555: 'fade rates', do you mean decay rates?

Yes, we changed this to decay rates

L558: 'Similarly,...' It is unclear what is referred back to with the similarly. Add what is being compared here, e.g., 'similarly to the radio light curves...' and adjust by what is actually referred.

- **We replace with 'Meanwhile,...'**

L566-569: This sounds like an important conclusion and should be discussed in the discussion, e.g., show a figure to compare the LFBOT radio behaviour to the CCSNe behaviour and discuss what the implications of different behaviour is.

- In figure 4, we add a dashed line showing the division between likely CSM interaction and the regular SNe radio regime. We also add sentences on how the diversity in CCSNe behavior implies that they can occur with and without a dense medium, but the LFBOT behavior indicates that the dense CSM is a consistent condition

L585-587: It would be nice to show the upper limits in a plot (or if there already is a plot, refer to it here) with examples of X-ray light curves of the other types of transients mentioned here, and AT2018cow.

- **The Swift upper limits are shown in Figure 3, and we add the jetted TDE light curve of AT2022cmc**

L588-591: This statement needs to be expanded on, by referring back to figure 3 which shows this behaviour.

- **We add references to Figure 3**

Section 4

This section should be directly after the observations, and it should be split between the radio analysis and the host galaxy analysis. The host galaxy analysis could even go into the discussion perhaps. In Section 4.1 a lot of assumptions are lacking explanation/justification.

- **We move this section to be directly after the observations, and clarify the assumptions according to your other comments in this section.**

Section 4.1

L609-613: The assumption that the shocks are not relativistic should be justified or at least discussed in terms of how reasonable it is.

- **We mention that looking at the $\Gamma\beta$ returned by this model for a self-consistency check does show that some objects approach the trans-relativistic regime, so there will be some numerical error in our stated values.**

L614-617: Why can we assume equipartition between the three? Is there a physical reason for it or was it just done to be consistent with Soderberg+2010a?

- **It was done to be consistent with Soderberg+2010a, as well as previous LFBOT papers, which may consider a couple of sets of values for the epsilons but always consider a case where both are equal to $\frac{1}{3}$.**

L621-624: Was $f=0.5$ simply chosen to allow comparison with previous LFBOT work or is there a physical reason for it as well?

- **It was chosen simply for comparison with other works. The filling factor affects the overall scale of the modeled flux given a set of values for the physical parameters**

L624-628: Is this typical value of the power-law index found through fitting to the radio spectra or through a different way?

- **Added clarification that previous estimates were done by fitting to the slope of the self-absorbed half of the radio SED, and that we don't quite have enough points to get a confident slope estimate so we just use a fiducial value. It is true that there are inconsistencies between our observed non-self-absorbed slope and that required by Chevalier's formulation, which we mention in the discussion.**

Equations (2)-(6): This might just be ignorance on my part, but if you assume equipartitions of the different energies, why are ϵ_e and ϵ_B not filled in in the equations? Because you know the value for them.

- **In our judgement, including the variables allows readers who prefer a different choice for these variables to easily recalculate our estimates for the parameters**

L644: Is the flux also in the frame of the transient?

- **Yes, we restructure the sentence to clarify both the values for the peak frequency and flux at peak frequency are the rest frame values**

L660-661: Why the smoothed broken power-law? Is this the standard in the field or is there another reason for it?

L663: Again, was $s=1$ chosen as in previous work to be consistent and allow for comparison or did previous work give a physical reason for why this is the best value for s ?

- **For the smoothed broken power-law and choice of $s=1$, it was chosen to be consistent with previous work.**

L669-670: "we impose a 10% error floor on the flux density" this needs more explanation, currently it is unclear what this means.

- **We add a sentence describing the an extra error equal to 10% of the flux density is applied in quadrature to each observation to account for differences in observatory systematics**

L674-678: How does using the detection closest to the inferred peak change if this point is just before or just after the inferred peak?

- **This affects whether the parameter limits we derive are upper or lower limits, as seen in the synchrotron parameters table in the Appendix. If the closest detection is at a lower frequency than the peak (in the self-absorbed regime), then we assume a basic scaling of $F \sim \nu$ for the SED up to the peak, and plug this into the equations to determine which direction the limits would be, knowing that the peak must have a higher value of F by definition. If the detection is at a higher frequency, we would use $F \sim \nu^{-1}$ instead. Sometimes, the direction is unaffected whether we use $F \sim \nu$ or $F \sim \nu^{-1}$, because the equation has a weak dependence on ν . We account for this by changing the directions of the arrows in Figure 8 and changing the directions of the various parameter limits in Table 8**

L683-687: "We find... 10^{17} cm." The implication of this results needs to be discussed in more detail in the discussion

- **In the discussion, we mention that this sets a timescale for how long ago the dense medium was created if it was mass emitted by the LFBOT's progenitor**

Section 4.2

L690: "their" is plural, but 'the expanded sample or LFBOTs' is singular. Suggested sentence: ".LFBOTs, we also examine the distribution of offsets from the host galaxy nuclei" or something equivalent

- **We fix the grammatical error with a similar sentence as to what was suggested**

L695-698: Why is radio the best wavelength to determine the position of the transients?

- **For each object that we use the radio localization for, we have at least one VLA or NOEMA observation that is better than an arcsecond, and often to even better resolution than that if the observation was taken in a higher resolution configuration of the VLA. For more details, we do provide the VLA configuration in Table 5, so that a reader could look at the typical synthesized beamwidths for that configuration.**

L708: The sample size was still doubled, so even though the numbers are small, you have a better idea than previous works, that can be emphasised here!

- **We a short sentence at the start of this section (now 4.1) taking note of this**

Section 4.3

A short explanation of why Prospector was chosen as the SED fitting code of choice over for example BAGPIPES should be added. First describe the fitting procedure, then list the table and figures the results can be found in at the end.

- **We mainly use prospector to easily compare with the host galaxy fitting done in previous LFBOT host galaxy papers, like Chrimes et al. 2024, Ho et al. 2023, Yao et al. 2022. We restructured the order of the descriptions.**

Figure 9: If the median parameters are used for the analysis in the rest of the paper, why not plot them here instead of the maximum-likelihood parameters?

- **The median is calculated individually parameter by parameter, which means that each of the median values may be coming from different models, so that there may not be a single model that has all of the median parameters that we can plot. We also change the choice of maximum-likelihood model to the maximum a posteriori model, which should be very similar in practice**

L722-723: This sentence currently states the photometry was fit to the model, this should be the other way around. A sentence on what dynesty is and what it does should also be added.

- **We fix the sentence, and mention that dynesty runs dynamic nested sampling for Bayesian inference**

L728: Justify why this form of the SFH was chosen.

- **This falls under our explanation that we choose parameters to mimic previous LFBOT papers.**

L744-745: How does the bias introduced here influence the results?

- **This could bias our sample host masses and star-formation rates to a higher value. Metallicity could be affected in an unclear way via the mass-metallicity relation, but some of our hosts are in regimes where metallicity is roughly constant with mass. We add a mention of this effect on star-formation rate.**

L749: Is it a realistic assumption to assume this sample of 150 CCSNe host galaxies as representative for the true CCSNe host sample? This needs to be discussed.

- **We prioritized having a sample that is unbiased. While there would be some inaccuracy from a small sample size, most of the error will be from our smaller LFBOT sample size.**

L756-760: What is the effect of not taking into account systematic differences between galaxy samples at the different redshifts? There are known differences between galaxy samples at different redshifts, such as their masses and star formation rates, so it is possible to quantify how this influences the results/conclusions.

- **While there are systematic differences, the change in most parameters from $z \sim 0.1$ to $z \sim 0.3$ would be smaller than other sources of error (like the uncertainties in the prospector fitting procedure). The change in star-formation rate can be significant, but we only compare galaxy masses between samples in this paper.**

L775: Are alpha and p different? Or should all of these be the same (either alpha or p)?

- **Apologies, p should be p-value. We reworded this sentence to be more explicit that they are different. For a full explanation: In this case, the p-value is a measure of how different the two observed distributions we are testing are. A p-value near zero means the distributions are very different, while a p near one means they are almost identical. Alpha (the significance level) is the threshold that the p-value must be less than to accept that there exists a statistically significant result in our analysis. It is also equal to the probability that we accept a false positive result. So, we chose $\alpha=0.05$ to establish a standard for how different the observed distributions have to conclude that the underlying distributions are different. The p-value is calculated from the two observed distributions we are comparing, and in each case the p-value was greater than 0.05, so we could not statistically conclude that the distributions are significantly different.**

Table 7 in appendix B: The photometry values are missing uncertainties, which should be present here. Additionally, it would be useful to indicate with an asterisk or equivalent which sources were outside the SDSS footprint and therefore have the Pan-STARRS luminosities listed, for clarity.

- **We add errors, and clarify that it is AT2023vth that has Pan-STARRS values**

Section 5

In general, this section contains some good discussion of some of the results. What should be added here is the effect of the assumptions made in previous sections, reflecting on how they influence the results.

Section 5.1

L792-795: This statement would benefit from saying why the consistent radio properties mean a TDE scenario is likely (e.g., because all disruptions are different due to geometry of the system and will therefore create different configurations of material and reference this properly). It is also not clear from the sentence if ', as discussed in previous work ..' means these works discussed that TDEs were a likely explanation or if they also argued the TDE scenario is unlikely.

- **Reworded the sentences and explained the reasoning mentioned in Margutti et al. arguing against the TDE scenario**

L803-804: Why does a model involving a merger naturally explain the non-varied radio behaviour? Adding a sentence explaining what this is (even though you have referenced works showing this) would support this claim.

- **The merger papers can calculate the expected circumburst density from merger scenarios and found that they are consistent with estimates from LFBOT papers. These expected densities should also hold across many instances of the same merger scenario, producing similar medium densities, which would explain our non-varied radio characteristics. We add a couple sentences explaining this,**

L805: The discussion of the peculiar case of AT2024aehp should be a separate section in the discussion, which discusses all the peculiar behaviour in one go so it's nicely located in one bit of the paper. It should also include a bit of discussion about if this is the only source within the LFBOTs that has a different progenitor, and how leaving this source out of the larger sample would influence the results.

- **We reorganize the text to include the discussion on AT2024aehp into a single subsection, including some text from the observational comparison section.**

L822: 'one of the closest' is too vague, please quantify

- **We change this wording to AT2024aehp being one of two LFBOTs that have localizations that are consistent with being nuclear**

L813: claiming the properties are similar to those of TDEs needs to be supported by references to papers about TDE properties (e.g. Charalampopoulos P., et al., 2022, A&A, 659, A34 for spectroscopic properties)

- **Our claim that AT2024aehp is similar to TDEs is based on the slower light curve decay, late radio brightening, and its nuclear position in the host. The light curve decay and radio brightening are mentioned and cited in next sentences, and we add a sentence and citation to the position**

Figure 12: From the caption is it unclear why there is a fit of a $t^{-5/3}$ to this lightcurve. Mentioning this is the expected decay of a TDE light curve for comparison will make the point of the figure clearer.

- **Added this to the Figure 12 caption**

Section 5.2

This section feels a bit out of place, because the paper up until now is not necessarily focused on the observing strategy. It is absolutely useful to talk about future considerations, but it would feel more at home in the paper if observing/follow-up strategy was discussed more before now. In most of this section it is stated that more observations are needed, but without explaining why those observations are needed. What will those extra observations bring us scientifically?

- **We add explanations of the scientific rationale behind the additional recommended observations. We add more detail to the follow-up strategy used in the paper earlier, at around line 150**

L832: This section talks about the observing strategy being influenced by the one peculiar LFBOT that was observed in this paper. Do we need to adjust our observing strategy to catch more of these peculiar ones? It feels a bit strange to change observing strategy based on one outlier, so this needs to be clarified.

- **We reworded some sentences to better catch the intent of these suggestions: we don't necessarily expect another AT2024aehp-like LFBOT, but it would be good to rule out some of its characteristics from future LFBOTs to make sure it really is unique. For example, it would be helpful if we later time photometry for some of the early LFBOTs, just to better characterize the fading of the light curve**

L847: Why do we need more Chandra observations? What will it bring us?

- **X-ray detections can be compared to UV/optical flux for to estimate the optical depth of the ejecta to the X-rays, X-rays can also probe the regions closest to the LFBOT remnant/compact object which we don't get from most other bands**

L860: How big are the uncertainties introduced by the non-relativistic assumptions? Is this big enough to influence the results or conclusions?

- **Using the estimates in Ferguson & Margalit 2025 that values may differ by an order of magnitude when you get to $\gamma\beta \sim 1$. Since our conclusions rely on similarity between the different LFBOTs, those claims wouldn't be affected. However, the values may indeed be influenced. We have added sentences explaining this at the end of the paragraph.**

L865-866: Not doing this for the analysis presented in this work needs to be justified.

L877-879: If these models already exist, why were they not used in this work?

- **This paper presents new data for LFBOTs and compares them to past LFBOTs using previously used techniques. Applying new models to the data is beyond the scope of the work and will be left for future analysis. Some of the new models also require some further development to be readily applied to radio SED observations. We have added an additional sentence to this section clarifying that.**

L890: How would using models with a non-parametric SFH change the results and conclusions?

- **A test model run with a basic non-parametric SFH did change the mass and sfr by 1-2 sigma. We add text describing this. We do not report the non-parameteric SFH results. We aimed to present results that can most easily be compared with past papers of our LFBOTs. As stated before, we are applying previously used techniques and are leaving the applications of new or more complex techniques to future work. A recent paper did a similar host-galaxy analysis, but used a non-parametric SFH and also galaxy spectroscopy when available. The mass and SFR estimates are similar for AT2023fhn and AT2024qfm, but they find significantly higher galaxy masses for AT2022abfc and AT2023hkw. We add a sentence mentioning this with a citation (Nugent et al. 2026).**

L895-900: A discussion on the reliability of the method used to calculate the metallicity and stellar mass of the galaxies should be included. Fixing one to calculate the other and then vice versa warrants some justification of why this isn't expected to produce biased results.

- **This would indeed bias our metallicity estimates to more moderate values. We have added some sentences explaining this.**

L912-918: From this it reads like the results regarding the offsets should be taken with a grain of salt because of the lack of correction for bias. If that is indeed true, write it down.

- **This is true, so we have added the above wording.**

L934: A short explanation of how the above mentioned instruments/surveys will enable LFBOTs to be identified and followed-up earlier in their evolutions should be added. This could include increased cadence, bigger FoV or whatever else is relevant in this case.

- **We add some more details of how LSST and ULTRASAT would benefit LFBOT studies**

Section 6

This section should also be written in present tense. It contains the right information, but the impact of the work should be emphasized more in this section. Including more of the impact of the field will make it clearer why the work is important.

L949: Should this be sixth instead of fifth event?

- **We add a mention of AT2022abfc, which should be the fifth event we describe in the sequence, leaving AT2024aehp as the sixth.**

L974: re-brightening instead of brightening?

- **We believe that brightening is more appropriate, as both AT2024aehp and the TDEs we cite in the literature had a single brightening phase in the radio past a hundred days after the initial event, instead of an initial brightening, a plateau or fade, and then another period of brightening.**

L1008: What would being able to identify and follow-up LFBOTs earlier mean for the field? How will it advance the knowledge about LFBOT progenitors?

- **We add a statement on how earlier radio observations probe smaller shock radii and therefore more recently pre-explosion. An earlier UV discovery means a UV light curve on the rise which we haven't had. UV measurements can improve our blackbody fits and get better estimates of the photosphere's effective temperature and radius (Pursiainen et al. 2025). Measuring the rising UV light curve would also help see early dust NIR (Metzger & Perley 2021). These statements were also moved to the end of the Discussion section instead of being in the Summary**

Appendix A

The classification of transients as LFBOTs is not done based on the same criteria for all sources, e.g., colour is not used for all sources. Why is this and why can we still classify these events as LFBOTs with a part of the selection criteria instead of all of them? For some of the observations mentioned, the PI is listed and for others the PI is not listed, this should be consistent for all of the observations if they stay included.

- **We have removed the PI footnotes unless the observation is not mentioned in a citable astronote, in order to give some reference to which observation we mention.**
- **For AT2023hkw and AT2024qfm, we do not mention color as they were only found to be blue a little bit after the initial Astronotes that motivated follow-up.**
- **For AT2024aehp, we add the mention of blue colors that was present in the discovery astronote, but was previously omitted in this paper**
- **AT2022abfc is a tricky case, because there is not enough photometry to confirm a blue color. We classify it as an LFBOT as it meets the other criteria, but this is one of the reasons why we obtained more sensitive follow-up photometry in the later LFBOTs.**

The discovery procedure in the ZTF data as described in Appendix A also includes a description of follow-up observations triggered by the authors, but not all of them. Either the title of Appendix A should be changed to state it includes a description of follow-up observations triggered and the full list of observations should be mentioned, or the mention of follow-up observations should be removed from this section entirely. This is also one of the sections that mentions a lot of other papers being written with data on each of these six sources, which is therefore not included in this manuscript. These gaps in the data included in the manuscript make it so the dataset presented is not complete, and I wonder if this influences the results and conclusions of the paper.

- The aim of this section is to describe the observations that led to the classification of each as an LFBOT. We edit the title and the first paragraph to reflect this. We only include the follow-up observations that contribute to each object's classification, such as initial X-ray or radio detections.
- The late-time VLT observations of AT2023vth are outside of this paper's scope, which is focused on early-time observations and some later radio data. It is true that AT2024qfm's radio observations could change the overall picture, but as our conclusions on the radio observations are drawn using observations of four of the other five LFBOTs (excepting AT2024aehp, which is unique) and much of the previous LFBOT sample, we still have confidence in their validity.